UPSC EPFO · APFC RECRUITMENT TEST · MODULE 8 · CHAPTER F8.6

Mensuration and Geometry

Chapter F8.6 — The Chapter the 2023 Paper Drew From Most Heavily

Module	**Foundation Layer — Module 8 — English, Quantitative Aptitude & Reasoning**
Chapter	**Chapter F8.6**
Topic	**Official syllabus head 7 — Elementary Mathematics and Statistics**
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HIGH YIELD · FOUR OF THE FIFTEEN QUANT ITEMS CAME FROM THIS CHAPTER

Blueprint §4.4's record of the 2023 Quant block names, among fifteen questions:

- **the mensuration of a hemisphere**
- **a melted disc recast into another solid**
- **circle geometry**
- **heights and distances**

That is four questions — 10 marks — from one chapter, and no other Quant chapter supplies as many. If you prepare one chapter in the Quant half properly, prepare this one.

It is also the most **deterministic** material in Module 8. A hemisphere's curved surface area is $2\pi r^2$ and nothing about the examiner's mood can change it. Unlike vocabulary, where knowledge is probabilistic, a formula known is a mark banked — provided you know **which** of a matched pair is being asked, which is where §2.2 concentrates.

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0. How This Chapter Works

STUDY SESSION · CHAPTER F8.6

1. **Session 1** — The two-dimensional formula sheet (§1) ★★★★★
2. **Session 2** — Solids, and the hemisphere in particular (§2) ★★★★★
3. **Session 3** — Melting, recasting and volume conservation (§3) ★★★★★
4. **Session 4** — Circle geometry (§4) ★★★★★
5. **Session 5** — Triangles, Pythagoras and similarity (§5) ★★★★☆
6. **Session 6** — Heights and distances (§6) ★★★★☆

Then 30 graded questions in §7. F8.0's calendar takes §7 Q1–10 on Tuesday of week 2, Q11–20 on Thursday of week 2 and Q21–30 on Thursday of week 4, then works the errors in week 6 and takes all 30 timed in week 9.

MUST MASTER · HOW TO LEARN A FORMULA SHEET SO IT SURVIVES THE EXAM HALL

Do not read these tables. **Reproduce them.** The difference decides the marks.

1. **Write the sheet out from memory once a week**, on one side of paper. The first attempt will have gaps; the fourth will not
2. **Learn the matched pairs together, never separately** — curved against total surface area, sphere against hemisphere, cone against cylinder. The examiner's only real weapon in mensuration is offering the **other** member of the pair, correctly computed. F8.0 §5 predicts this will be a leading entry in your error log, and Blueprint §8 names it as a trap pattern across the manual
3. **Take π = 22/7 whenever the radius is a multiple of 7**, which in an examination it almost always is. That is the setter telling you which value to use
4. **Check the dimension of your answer.** Area answers are in cm^2, volumes in cm^3. An answer with the wrong dimension is wrong even if the arithmetic is right, and the wrong-dimension option is always on the paper

1. Session 1 — Plane Figures ★★★★★

1.1 The area sheet

Figure	Area	Perimeter
Square (side a)	**a^2**	4a · **diagonal = a root 2**
Rectangle	**lb**	2(l + b) · **diagonal = root($l^2 + b^2$)**
Triangle	**½ × base × height**	sum of the sides
Triangle, three sides known	**Heron:** root[s(s-a)(s-b)(s-c)], s = (a+b+c)/2	
Equilateral triangle (side a)	**(root 3 / 4) a^2**	3a · **height = (root 3 / 2) a**
Right triangle	½ × the two legs	
Parallelogram	**base × height**	2(a + b)
Rhombus	**½ d_1d_2**	4a
Trapezium	**½ (a + b) × h** — half the sum of the parallel sides times the height	
Circle	**πr^2**	**circumference $2\pi r$**
Semicircle	$\frac{1}{2}\pi r^2$	**perimeter $\pi r + 2r$** — the arc **and the diameter**
Sector, angle θ	$(\theta/360) \times \pi r^2$	arc = $(\theta/360) \times 2\pi r$
Ring (annulus)	$\pi(R^2 - r^2)$	

TRAP · THE PERIMETER OF A SEMICIRCLE IS NOT HALF THE CIRCUMFERENCE

Half of $2\pi r$ is πr, and πr is only the **curved** part. A semicircle is bounded by the arc **and by the diameter**, so its perimeter is **$\pi r + 2r$**.

For r = 7 that is 22 + 14 = **36 cm**, not 22 cm — and 22 is on the paper as an option. Q6.

The same logic governs a sector: its perimeter includes the **two radii**. Wherever a figure is made by cutting, ask what new edge the cut created.

1.2 The scaling rule

MUST MASTER · LENGTHS SCALE BY K, AREAS BY K^2, VOLUMES BY K^3

If every length in a figure is multiplied by k, then **areas multiply by k^2** and **volumes by k^3**.

Everything in the following list is the same fact:

- A radius increased by 20% increases the area by **44%**, because $1.2^2 = 1.44$ — Q13
- Similar triangles with sides 3 : 5 have areas **9 : 25** — Q20
- Doubling the radius of a sphere multiplies its volume by **eight**
- A cube whose edge is halved has **one quarter** the surface area and **one eighth** the volume

Never apply a percentage change directly to an area. Convert it to a length factor, square it, then convert back. This is F8.5 §1.3's moving-base point in geometric dress, and it is the single most frequently examined idea in mensuration after the formula sheet itself.

CHECK YOURSELF · SESSION 1

1. Write the area sheet from memory
2. Perimeter of a semicircle of radius 14 cm?
3. Area of an equilateral triangle of side 6 cm, and its height?
4. If a radius falls by 10%, what happens to the area?
5. Areas of two similar triangles with sides in the ratio 2 : 7?

2. Session 2 — Solids ★★★★★

2.1 The solid sheet

Solid	Volume	Curved / lateral surface	Total surface
Cube (edge a)	**a^3**	$4a^2$	**$6a^2$** · diagonal **a root 3**
Cuboid	**lbh**	2h(l + b)	**2(lb + bh + hl)** · diagonal root($l^2+b^2+h^2$)
Cylinder	**$\pi r^2 h$**	**$2\pi rh$**	**$2\pi r(r + h)$**
Cone	**$\frac{1}{3}\pi r^2 h$**	**πrl**	**$\pi r(r + l)$ · $l^2 = r^2 + h^2$**
Sphere	**$\frac{4}{3}\pi r^3$**	—	**$4\pi r^2$**
Hemisphere	**$\frac{2}{3}\pi r^3$**	**$2\pi r^2$**	**$3\pi r^2$**
Frustum	$\frac{1}{3}\pi h(R^2 + r^2 + Rr)$	$\pi l(R + r)$	

2.2 The hemisphere — the 2023 item, and the trap inside it

MUST MASTER · A HEMISPHERE HAS THREE DIFFERENT SURFACE AREAS, AND THE QUESTION PICKS ONE

This is the matched pair that the 2023 paper tested, and the reason candidates lose the mark is not the formula but the **reading**.

What is asked	Formula	For r = 7
Curved surface area	**$2\pi r^2$**	**308 cm^2**
The flat circular face alone	πr^2	154 cm^2
Total surface area	**$3\pi r^2$**	**462 cm^2**

The total is $3\pi r^2$ because it is the curved $2\pi r^2$ PLUS the flat circular base πr^2. That is the whole derivation, and if you remember the reason you cannot confuse the two.

When is it $2\pi r^2$ and when $3\pi r^2$? Ask whether the flat face is part of the surface in question:

- A **hemispherical bowl or dome** — you are painting or wetting only the curved part » **$2\pi r^2$**
- A **solid hemisphere** resting on its base, to be painted all over » **$3\pi r^2$**
- Words like *curved*, *dome*, *external* point to $2\pi r^2$; *total*, *whole*, *solid* point to $3\pi r^2$

Q2 and Q3 are the same hemisphere asked twice, deliberately, so that the pair is drilled together as §0 requires.

2.3 The cone's slant height

$l^2 = r^2 + h^2$, so the slant height is always the largest of the three. A question that gives you the slant height and asks for **volume** is a two-step question: find h by Pythagoras first, then use $\frac{1}{3}\pi r^2 h$. For r = 6 and l = 10, h = 8 and V = 96π — Q30.

The reverse also holds: a question giving r and h and asking for **curved surface area** needs l first. The examiner offers the answer obtained by using h in place of l.

2.4 The ratio worth memorising

KNOW · CONE : HEMISPHERE : CYLINDER = 1 : 2 : 3

On the **same base and of the same height** — so h = r — the three volumes are:

- Cone $\frac{1}{3}\pi r^2 \cdot r$ = $\mathbf{\frac{1}{3}\pi r^3}$
- Hemisphere $\mathbf{\frac{2}{3}\pi r^3}$
- Cylinder $\pi r^2 \cdot r$ = $\mathbf{\pi r^3}$

which is **1 : 2 : 3**. Q22.

It is worth memorising as a fact because it appears as a question in its own right, and because it is a free sanity check: a cone is always **a third** of the cylinder that encloses it, and a sphere **two thirds** of it.

CHECK YOURSELF · SESSION 2

1. Write the solid sheet from memory
2. Give all three surface areas of a hemisphere of radius 7 cm, and say when each applies
3. Volume of a cone of radius 3 cm and slant height 5 cm?
4. Ratio of the volumes of a cone, hemisphere and cylinder on the same base with h = r?

3. Session 3 — Melting and Recasting ★★★★★

3.1 The one principle

> **When a solid is melted, or a wire is bent, the VOLUME is conserved. The surface area is not.**

Every question in this family — and the 2023 paper's **melted disc** is one — is solved by writing one equation:

> **volume of the original = volume of the new shape**

and solving for the unknown dimension. Nothing else is involved.

3.2 The three shapes it takes

Shape of question	Equation
One solid into another	$V_1 = V_2$, solve for the new dimension
One solid into many small ones	**number = V_1 ÷ V_small**
A wire bent into a new figure	**length is conserved** — perimeter to perimeter, not area to area

> **Worked — the 2023 melted-disc shape.** A metallic circular disc of radius 12 cm and thickness 2 cm is melted and recast as a sphere. The disc is a cylinder: $V = \pi \times 144 \times 2 =$ **288π**. Then $\frac{4}{3}\pi r^3 = 288\pi$ gives $r^3 = 216$ and **$r = 6$ cm** — Q12.
>
> **Worked — into many.** A hemispherical bowl of internal radius 9 cm is emptied into cylindrical bottles of radius 1.5 cm and height 4 cm. Bowl = $\frac{2}{3}\pi \times 729 = 486\pi$; bottle = $\pi \times 2.25 \times 4 = 9\pi$; so **54 bottles** — Q21.
>
> **Worked — a wire.** A wire bent into a circle of radius 7 cm is re-bent into a square. The **length** is conserved: $2\pi \times 7 = 44$ cm, so the side is **11 cm** — Q29. The areas differ (154 against 121), and that difference is itself sometimes the question.

TRAP · WHAT IS CONSERVED DEPENDS ON WHAT IS BEING RESHAPED

- **A solid melted** » **volume** is conserved. Its surface area changes, usually falling, because a sphere

is the most economical shape

- **A wire or a rope bent** » **length** is conserved, so you work with perimeters
- **A sheet folded into a cylinder** » **area** is conserved, and it becomes the curved surface area

Getting this wrong produces a clean, confident, wrong answer, so name the conserved quantity **in writing** before setting up the equation. In Q29 the option 14 cm is waiting for the candidate who conserved area instead of length.

CHECK YOURSELF · SESSION 3

1. What is conserved when a solid is melted? When a wire is bent?
2. A sphere of radius 6 cm is recast as a cylinder of radius 4 cm. Height?
3. How many cones of radius 3.5 cm and height 3 cm come from a sphere of radius 10.5 cm?
4. Why does surface area fall when scrap metal is melted into a ball?

4. Session 4 — Circle Geometry ★★★★★

The 2023 paper tested this head directly. The results below are the complete examinable set.

4.1 Angles

Result	Statement
Angle at the centre	**Twice** the angle subtended by the same arc at any point on the remaining circumference
Angle in a semicircle	**90°** — the angle subtended by a **diameter** at the circumference is a right angle
Same segment	Angles in the same segment are **equal**
Cyclic quadrilateral	**Opposite angles are supplementary** — they add to 180°
Exterior angle	The exterior angle of a cyclic quadrilateral equals the **interior opposite** angle
Alternate segment	The angle between a tangent and a chord equals the angle in the alternate segment

4.2 Chords and tangents

Result	Statement
Perpendicular from the centre	**Bisects the chord** — so radius, half-chord and distance form a right triangle
Equal chords	Are **equidistant** from the centre, and subtend equal angles at it
Tangent and radius	Meet at **90°** at the point of contact
Two tangents from a point	Are **equal** in length, and subtend equal angles at the centre
Tangent–secant	$PT^2 = PA \times PB$
Intersecting chords	$AP \times PB = CP \times PD$

MUST MASTER · THE RIGHT TRIANGLE HIDDEN IN EVERY CHORD AND TANGENT QUESTION

Nearly every numerical circle question in this examination is a **Pythagoras question in disguise**, and the two configurations are these:

1. A chord. Drop a perpendicular from the centre. It bisects the chord, giving a right triangle with the **radius as hypotenuse**, half the chord, and the distance from the centre:

> **r^2 = (half the chord)^2 + (distance from the centre)^2**
>
> A chord of 16 cm at 6 cm from the centre: $r^2 = 8^2 + 6^2 = 100$, so **r = 10 cm** — Q14.

2. A tangent. The tangent meets the radius at 90°, so for an external point P at distance d:

> **(tangent length)^2 = $d^2 - r^2$**
>
> From a point 13 cm from the centre of a circle of radius 5 cm: 169 - 25 = 144, so the tangent is **12 cm** — Q16.

Both are the 3-4-5 family in disguise — 6-8-10 and 5-12-13. Recognising the triple removes the arithmetic entirely, which is how these come inside the 75–90 second budget.

4.3 The two-step angle chase

Circle questions at RT standard rarely give you the angle you need directly. They give an angle two steps away, and the route is nearly always: **isosceles triangle » angle at centre » angle at circumference**.

> **Worked.** O is the centre; A and B are on the circle; the angle OAB is 25°. Find the angle ACB, where C is on the major arc.
>
> **Step 1.** OA and OB are both radii, so triangle OAB is **isosceles** and the angle OBA is also 25°. **Step 2.** The angles of the triangle sum to 180°, so the angle **AOB = 130°**. **Step 3.** The angle at the circumference is **half** the angle at the centre: **ACB = 65°** — Q26.

Every radius drawn in a figure creates an isosceles triangle. **That is the move to look for first** when a question seems to give too little.

CHECK YOURSELF · SESSION 4

1. State the six angle results and the six chord-and-tangent results
2. A chord of 24 cm lies 5 cm from the centre. Find the radius
3. Tangent length from a point 17 cm from the centre of a circle of radius 8 cm?
4. In a cyclic quadrilateral one angle is 85°. What is the opposite angle?
5. Why does every radius in a figure create an isosceles triangle?

5. Session 5 — Triangles ★★★★☆

5.1 Pythagoras, and the triples worth recognising

> **hypotenuse^2 = the sum of the squares of the other two sides**

Memorise these, because recognising one saves the whole computation:

3-4-5 · 5-12-13 · 8-15-17 · 7-24-25 · 9-40-41 — and every multiple, so **6-8-10**, **9-12-15**, **10-24-26**, **15-36-39**.

A ladder 13 m long with its foot 5 m from a wall reaches **12 m** up it — no arithmetic required, because 5-12-13 is on the list. Q17.

5.2 Similarity

Two triangles are similar if their angles are equal (AAA), or two sides are proportional with the included angle equal (SAS), or all three sides are proportional (SSS).

In similar triangles	Ratio
Corresponding sides, heights, medians, perimeters	**k**
Areas	**k^2**
Volumes of similar solids	**k^3**

5.3 The other results that get asked

Result	Statement
Angle sum	180° in a triangle · **exterior angle = sum of the two interior opposite angles**
Midpoint theorem	The segment joining the midpoints of two sides is **parallel to the third and half its length**
Centroid	Medians meet at the centroid, which divides each median **2 : 1** from the vertex
Angle bisector	Divides the opposite side in the ratio of the adjacent sides
Inequality	Any two sides together exceed the third · the largest side faces the largest angle
Equilateral, side a	Height **(root 3 / 2) a** · area **(root 3 / 4) a^2**
30-60-90	Sides in the ratio **1 : root 3 : 2**
45-45-90	Sides in the ratio **1 : 1 : root 2**

KNOW · THE TWO SPECIAL TRIANGLES CARRY THE HEIGHTS-AND-DISTANCES QUESTIONS

The **30-60-90** ratio 1 : root 3 : 2 and the **45-45-90** ratio 1 : 1 : root 2 are why every heights-and-distances question in this examination uses 30°, 45° or 60° and never 37°.

Learn them as *shapes*, not as trigonometry: in a 30-60-90 triangle the side opposite 30° is **half** the hypotenuse, and the side opposite 60° is that half **times root 3**. §6 is an application of these two triangles and nothing more.

CHECK YOURSELF · SESSION 5

1. List five Pythagorean triples
2. Areas of similar triangles whose perimeters are in the ratio 4 : 9?
3. In what ratio does the centroid divide a median?
4. Give the side ratios of the 30-60-90 and 45-45-90 triangles

6. Session 6 — Heights and Distances ★★★★★

6.1 The three values you need

θ	sin	cos	tan
30°	½	(root 3)/2	**1 / root 3**
45°	(root 2)/2	(root 2)/2	**1**
60°	(root 3)/2	½	**root 3**

In practice you need only the tangents, because a height-and-distance problem relates a vertical to a horizontal:

> **tan(angle of elevation) = height ÷ horizontal distance**

6.2 The three configurations

Given	Method
Distance and angle, find height	height = distance × tan θ
Height and angle, find distance	distance = height ÷ tan θ
Two angles from two points on the same line	Write both distances in terms of h, and **subtract**

> **Worked — the single-angle case.** From 30 m from the foot of a tower the elevation is 45°. Since tan 45° = 1, the height **equals** the distance: **30 m** — Q18. At 30° from 90 m: h = 90/root 3 = **30 root 3**, about 52 m — Q19.
>
> **Worked — the two-angle case.** The elevation changes from 30° to 60° as an observer walks 40 m towards the tower. From the far point the distance is h/tan 30° = h root 3; from the near point h/tan 60° = h/root 3. The **difference is the 40 m walked**:
>
> h root 3 - h/root 3 = 40, so h(3 - 1)/root 3 = 40, giving 2h = 40 root 3 and **h = 20 root 3**, about 34.6 m — Q27.

MUST MASTER · THE ANGLE OF DEPRESSION EQUALS THE ANGLE OF ELEVATION

The angle of **elevation** is measured from the horizontal **upwards** to the line of sight; the angle of **depression** from the horizontal **downwards**. Because the two horizontals are parallel, **the angle of depression from the top of a tower to a point equals the angle of elevation of the top from that point.**

So a question phrased in depressions can be redrawn as one in elevations, which is the form you have practised. That is the only thing you need to know about depression.

Two further habits. First, **draw the triangle** — every error in this topic is a mislabelled diagram rather than a failed computation. Second, note that **root 3 is about 1.732 and root 2 about 1.414**, so you can eliminate options by rough size: 20 root 3 is about 34.6, which instantly separates it from 40 root 3 at about 69.

CHECK YOURSELF · SESSION 6

1. Give the tangents of 30°, 45° and 60°
2. At what distance does a 60 m tower subtend 45°? And 30°?
3. State the relation between the angles of elevation and depression, and why
4. Two elevations 45° and 60°, observer walks 20 m. Set up the equation

7. Graded Practice — 30 Questions ★★★★★

Target: 75 to 90 seconds each (F8.0 §1.1). Take **π = 22/7** unless told otherwise. Draw the figure for every geometry question — it costs ten seconds and prevents most errors.

7.1 Level 1 — Formula recall

Q1. The circumference of a circle is 44 cm. Its area is (a) 144 cm^2 (b) 154 cm^2 (c) 176 cm^2 (d) 196 cm^2

Q2. The **curved** surface area of a hemisphere of radius 7 cm is (a) 154 cm^2 (b) 231 cm^2 (c) 294 cm^2 (d) 308 cm^2

Q3. The **total** surface area of a solid hemisphere of radius 7 cm is (a) 462 cm^2 (b) 308 cm^2 (c) 154 cm^2 (d) 616 cm^2

Q4. The volume of a cylinder of radius 7 cm and height 10 cm is (a) 440 cm^3 (b) 1,100 cm^3 (c) 1,540 cm^3 (d) 2,200 cm^3

Q5. The area of an equilateral triangle of side 8 cm is (a) 32 root 3 cm^2 (b) 16 root 3 cm^2 (c) 8 root 3 cm^2 (d) 64 root 3 cm^2

Q6. The perimeter of a semicircle of radius 7 cm is (a) 22 cm (b) 25 cm (c) 29 cm (d) 36 cm

Q7. The diagonal of a square of side 10 cm is (a) 10 root 2 cm (b) 20 cm (c) 10 root 3 cm (d) 5 root 2 cm

Q8. The volume of a sphere of radius 3 cm is (a) 12π cm^3 (b) 27π cm^3 (c) 36π cm^3 (d) 108π cm^3

Q9. The area of a trapezium whose parallel sides measure 12 cm and 20 cm and whose height is 5 cm is (a) 60 cm^2 (b) 80 cm^2 (c) 100 cm^2 (d) 160 cm^2

Q10. The angle subtended by a diameter at any point on the circumference of a circle is (a) 45° (b) 60° (c) 120° (d) 90°

7.2 Level 2 — Two steps

Q11. A sphere of radius 6 cm is melted and recast into a solid cylinder of radius 4 cm. The height of the cylinder is (a) 12 cm (b) 16 cm (c) 18 cm (d) 24 cm

Q12. A solid metallic circular disc of radius 12 cm and thickness 2 cm is melted and recast into a sphere. The radius of the sphere is (a) 6 cm (b) 8 cm (c) 9 cm (d) 12 cm

Q13. If the radius of a circle is increased by 20%, its area increases by (a) 20% (b) 44% (c) 40% (d) 21%

Q14. A chord of length 16 cm lies at a distance of 6 cm from the centre of a circle. The radius of the circle is (a) 8 cm (b) 12 cm (c) 14 cm (d) 10 cm

Q15. The angle between a tangent to a circle and the radius drawn to the point of contact is (a) 90° (b) 60° (c) 45° (d) dependent on the length of the tangent

Q16. The length of the tangent drawn from a point 13 cm from the centre of a circle of radius 5 cm is (a) 8 cm (b) 10 cm (c) 12 cm (d) 18 cm

Q17. A ladder 13 m long rests against a vertical wall with its foot 5 m from the wall. The height reached on the wall is (a) 10 m (b) 12 m (c) 8 m (d) 18 m

Q18. The angle of elevation of the top of a tower from a point on the ground 30 m from its foot is 45°. The height of the tower is (a) 15 m (b) 30 root 3 m (c) 10 root 3 m (d) 30 m

Q19. The angle of elevation of the top of a tower from a point 90 m from its foot is 30°. The height of the tower is (a) 45 m (b) 90 root 3 m (c) 30 root 3 m (d) 30 m

Q20. The corresponding sides of two similar triangles are in the ratio 3 : 5. The ratio of their areas is (a) 9 : 25 (b) 3 : 5 (c) 27 : 125 (d) 5 : 3

7.3 Level 3 — Recruitment Test standard

Q21. A hemispherical bowl of internal radius 9 cm is full of liquid, which is to be filled into cylindrical bottles of radius 1.5 cm and height 4 cm. The number of bottles required is (a) 27 (b) 54 (c) 62 (d) 72

Q22. A cone, a hemisphere and a cylinder stand on equal bases and have the same height. The ratio of their volumes is (a) 1 : 2 : 1 (b) 2 : 1 : 3 (c) 1 : 3 : 2 (d) 1 : 2 : 3

Q23. A metallic sphere of radius 10.5 cm is melted and recast into small cones each of radius 3.5 cm and height 3 cm. The number of cones so formed is (a) 126 (b) 63 (c) 108 (d) 252

Q24. In a cyclic quadrilateral ABCD, the angle A measures 70°. The angle C measures (a) 70° (b) 90° (c) 110° (d) 140°

Q25. Two chords AB and CD of a circle intersect at a point P inside the circle. If AP = 4 cm, PB = 9 cm and CP = 6 cm, then PD is (a) 4.5 cm (b) 6 cm (c) 8 cm (d) 13.5 cm

Q26. In a circle with centre O, the points A and B lie on the circle and the angle OAB measures 25°. If C is a point on the major arc, the angle ACB measures (a) 130° (b) 115° (c) 50° (d) 65°

Q27. As an observer walks 40 m towards a tower on level ground, the angle of elevation of its top changes from 30° to 60°. The height of the tower is (a) 40 root 3 m (b) 20 m (c) 20 root 3 m (d) 60 m

Q28. The area of the largest circle that can be inscribed in a square of side 14 cm is (a) 154 cm^2 (b) 196 cm^2 (c) 77 cm^2 (d) 616 cm^2

Q29. A wire in the form of a circle of radius 7 cm is bent into the form of a square. The side of the square is (a) 7 cm (b) 11 cm (c) 14 cm (d) 22 cm

Q30. The volume of a right circular cone of radius 6 cm and slant height 10 cm is (a) 120π cm^3 (b) 288π cm^3 (c) 80π cm^3 (d) 96π cm^3

8. Answers and Explanations

8.1 Level 1

Q	Ans	Explanation
1	b	$2\pi r = 44$ gives $r = 7$, so area = (22/7) × 49 = **154 cm²**. The radius being 7 is the setter's signal to use 22/7
2	d	**Curved** surface = $2\pi r^2$ = 2 × (22/7) × 49 = **308 cm²**. Note (a) is πr^2, the flat face alone
3	a	**Total** surface = $3\pi r^2$ = **462 cm²** — the curved 308 **plus** the flat 154. (b) is the curved area, the matched-pair trap of §2.2, and it is the option most often marked
4	c	$\pi r^2 h$ = (22/7) × 49 × 10 = **1,540 cm³**. (a) is the curved surface area, which is a dimension error — check cm^2 against cm^3
5	b	(root 3 / 4) × 64 = **16 root 3 cm²**, about 27.7. (d) omits the division by 4
6	d	$\pi r + 2r$ = 22 + 14 = **36 cm**. §1.1's trap: (a) is the arc alone and ignores the diameter
7	a	a root 2 = **10 root 2 cm**, about 14.1. (b) is the perimeter of two sides; (c) is the cube's diagonal formula misapplied
8	c	$\frac{4}{3}\pi$ × 27 = **36π cm³**. (d) is $4\pi r^3$, omitting the third
9	b	½ (12 + 20) × 5 = **80 cm²**. (d) forgets the half
10	d	**90°** — the angle in a semicircle. One of the six results in §4.1, and the most frequently used

8.2 Level 2

Q	Ans	Explanation
11	c	Volume conserved: $\frac{4}{3}\pi \times 216 = 288\pi$, and $\pi \times 16 \times h = 288\pi$, so h = **18 cm**
12	a	The disc is a cylinder: $\pi \times 144 \times 2 = 288\pi$. Then $\frac{4}{3}\pi r^3 = 288\pi$ gives $r^3 = 216$ and **r = 6 cm**. This is the 2023 melted-disc item; note the thickness is the cylinder's **height**
13	b	Length factor 1.2, so area factor $1.2^2 = 1.44$: an increase of **44%**. §1.2. (a) applies the percentage directly to the area, which is the standard error
14	d	The perpendicular from the centre bisects the chord: $r^2 = 8^2 + 6^2 = 100$, so **r = 10 cm**. The 6-8-10 triple
15	a	**90°**, always, at the point of contact — §4.2
16	c	$\text{Tangent}^2 = 13^2 - 5^2 = 144$, so **12 cm**. The 5-12-13 triple; (b) is 13 - 5 + 2, the answer of a candidate subtracting lengths instead of squares
17	b	$13^2 - 5^2 = 144$, so **12 m**. The same triple in an applied dress
18	d	tan 45° = 1, so the height **equals** the distance: **30 m**. The simplest configuration, and the one where candidates over-think and reach for root 3
19	c	h = 90 × tan 30° = 90/root 3 = **30 root 3 m**, about 52 m. (b) multiplies by root 3 instead of dividing — the commonest slip in the topic
20	a	Areas scale as the **square** of the side ratio: **9 : 25**. (b) is the side ratio and (c) the volume ratio, both offered

8.3 Level 3

Q	Ans	Explanation
21	b	Bowl = $\frac{2}{3}\pi \times 729 = 486\pi$; bottle = $\pi \times 2.25 \times 4 = 9\pi$; 486/9 = **54 bottles**. (a) is what you get from $\frac{1}{3}\pi$, i.e. treating the bowl as a cone
22	d	With h = r the volumes are $\frac{1}{3}\pi r^3$, $\frac{2}{3}\pi r^3$ and πr^3, so **1 : 2 : 3** — §2.4
23	a	Sphere = $\frac{4}{3}\pi \times 10.5^3 = 1{,}543.5\pi$; cone = $\frac{1}{3}\pi \times 12.25 \times 3 = 12.25\pi$; 1,543.5/12.25 = **126**. (b) halves it, which is what omitting the 4 in $\frac{4}{3}$ does
24	c	Opposite angles of a cyclic quadrilateral are **supplementary**: 180° - 70° = **110°**. (a) assumes they are equal, which is true only if both are 90°
25	b	Intersecting chords: AP × PB = CP × PD, so 4 × 9 = 6 × PD and **PD = 6 cm**. (d) is 36/... computed with the wrong pairing — note that the products pair each chord's **own** two segments
26	d	OA and OB are radii, so triangle OAB is isosceles and the angle OBA is also 25°; hence the angle AOB = 130°; hence the angle at the circumference is half, **65°**. §4.3's three-step chase. (a) is the central angle and (c) is 130 - 80, both offered as intermediate values
27	c	h root 3 - h/root 3 = 40 gives 2h/root 3 = 40 and **h = 20 root 3 m**, about 34.6. Rough-size check: the tower is under 40 m, which alone eliminates (a) at about 69 and (d)
28	a	The largest inscribed circle has diameter equal to the side, so r = 7 and area = **154 cm²**. (b) is the area of the square; (d) uses r = 14
29	b	**Length** is conserved when a wire is bent: $2\pi \times 7 = 44$ cm, so the side is 44/4 = **11 cm**. (c) conserves area instead — §3.2's trap
30	d	First h = root(100 - 36) = 8 by Pythagoras, then V = $\frac{1}{3}\pi \times 36 \times 8$ = **96π cm³**. (a) uses the slant height in place of the vertical height, which is the whole point of the question

THINK LIKE UPSC · THE FIVE SHAPES IN THIS BANK

1. **The matched pair, correctly computed.** Q2 against Q3 (curved against total), Q4(a), Q8(d), Q9(d), Q20(b) and (c), Q30(a). In every case the wrong option is a **real formula applied to the wrong question**. This is the dominant trap in mensuration and the reason §0 insists on learning pairs together
2. **The Pythagorean triple in disguise.** Q14, Q16, Q17, Q30 — 6-8-10 and 5-12-13, three times. Learn the five triples and these become recognition rather than computation
3. **What is conserved.** Q11, Q12, Q21, Q23, Q29. Volume for a melted solid, **length** for a bent wire. Naming the conserved quantity in writing is the whole method
4. **The scaling error.** Q13 and Q20. A percentage or ratio applied directly to an area instead of being squared. Lengths by k, areas by k^2, volumes by k^3
5. **The dimension check.** Q4 offers 440, which is the curved surface area in cm^2; Q28 offers 196, the area of the square. **Read the units of your own answer before you look at the options** — it is a two-second audit that catches a whole class of error

9. One-Minute Revision

ONE-MINUTE REVISION · CHAPTER F8.6

WEIGHT the heaviest Quant chapter — **hemisphere, melted disc, circle geometry and heights and distances all appeared in 2023** · **target 75–90 seconds** · take **π = 22/7** when the radius is a multiple of 7

PLANE square a^2, diagonal **a root 2** · rectangle lb, diagonal root(l^2+b^2) · triangle ½bh · **Heron root[s(s-a)(s-b)(s-c)]** · **equilateral (root 3/4)a^2, height (root 3/2)a** · parallelogram base × height · rhombus **½d_1d_2** · **trapezium ½(a+b)h** · circle **πr^2**, circumference $2\pi r$ · **semicircle perimeter πr + 2r, NOT πr** · sector (θ/360) of the circle · ring $\pi(R^2 - r^2)$

SOLIDS cube a^3, **$6a^2$**, diagonal a root 3 · cuboid lbh, 2(lb+bh+hl) · cylinder **πr^2h**, curved **$2\pi rh$**, total $2\pi r(r+h)$ · cone **⅓πr^2h**, curved **πrl, $l^2 = r^2 + h^2$** · sphere **$\frac{4}{3}\pi r^3$, $4\pi r^2$**

HEMISPHERE — THE 2023 ITEM volume **⅔πr^3** · **curved $2\pi r^2$** · flat face πr^2 · **total $3\pi r^2$ = curved + flat**. *Curved / dome / external* » $2\pi r^2$; *total / solid / whole* » $3\pi r^2$. For r = 7: **308 · 154 · 462**

CONE : HEMISPHERE : CYLINDER on the same base with h = r = 1 : 2 : 3

SCALING lengths **k**, areas **k^2**, volumes **k^3** · radius up 20% » area up **44%** · sides 3 : 5 » areas **9 : 25**

MELTING volume conserved, surface area not · into many: **number = V ÷ V_small** · **a bent wire conserves LENGTH** · a folded sheet conserves area

CIRCLE ANGLES centre = **twice** circumference · **semicircle = 90°** · same segment equal · **cyclic quadrilateral opposite angles supplementary** · exterior = interior opposite · alternate segment

CHORDS AND TANGENTS perpendicular from centre **bisects** the chord » **r^2 = (half chord)^2 + distance^2** · equal chords equidistant · **tangent meets radius at 90°** » **tangent^2 = $d^2 - r^2$** · two tangents from a point are equal · **PT^2 = PA × PB** · **AP × PB = CP × PD**

EVERY RADIUS MAKES AN ISOSCELES TRIANGLE — isosceles » angle at centre » halve it

TRIPLES 3-4-5 · 5-12-13 · 8-15-17 · 7-24-25 · 9-40-41 and multiples

TRIANGLES exterior = sum of interior opposite · midpoint segment is **half and parallel** · centroid **2 : 1** · **30-60-90 = 1 : root 3 : 2** · **45-45-90 = 1 : 1 : root 2**

HEIGHTS tan 30° = 1/root 3 · tan 45° = 1 · tan 60° = root 3 · height = distance × tan θ · **two angles: write both distances in h and subtract** · **depression = elevation** · **root 3 = 1.732, root 2 = 1.414** for size checks · **draw the triangle**

10. Five-Minute Wall Sheet

FIVE-MINUTE WALL SHEET · CHAPTER F8.6

HEMISPHERE: VOLUME $\frac{2}{3}\pi r^3$ · CURVED $2\pi r^2$ · TOTAL $3\pi r^2$ (= CURVED + FLAT)

$r = 7$ » 308 CURVED · 462 TOTAL. READ WHICH ONE IS ASKED

CONE $\frac{1}{3}\pi r^2 h$ · CURVED $\pi r l$ · $l^2 = r^2 + h^2$ — FIND l BEFORE ANY SURFACE AREA

SPHERE $\frac{4}{3}\pi r^3$ AND $4\pi r^2$

CONE : HEMISPHERE : CYLINDER = 1 : 2 : 3

SEMICIRCLE PERIMETER = $\pi r + 2r$. THE CUT MAKES A NEW EDGE

LENGTHS k · AREAS k^2 · VOLUMES k^3. RADIUS +20% » AREA +44%

MELTED » VOLUME CONSERVED. BENT WIRE » LENGTH CONSERVED

r^2 = (HALF CHORD)^2 + DISTANCE^2 · TANGENT^2 = $d^2 - r^2$

3-4-5 · 5-12-13 · 8-15-17 · 7-24-25

CENTRE ANGLE = TWICE THE CIRCUMFERENCE ANGLE · SEMICIRCLE = 90°

CYCLIC QUADRILATERAL: OPPOSITE ANGLES ADD TO 180°

AP × PB = CP × PD

RADIUS + RADIUS = ISOSCELES. THAT IS THE FIRST MOVE

tan 30 = 1/root 3 · tan 45 = 1 · tan 60 = root 3

TWO ANGLES, ONE WALK: h root 3 - h/root 3 = THE WALK

DEPRESSION = ELEVATION

CHECK THE UNITS: cm^2 OR cm^3?

ONE LINE FOR THE INTERVIEW Mensuration is the most reliable scoring block in an aptitude paper because the answers are determinate — which means the examiner's only real weapon is offering the other member of a matched pair, correctly computed, and the defence is to read whether the question said curved or total.

ACTION · NEXT

F8.7 — Algebra, Probability and the Number System. The 2023 paper took **quadratic roots** and a **probability** item from it.

Tonight's drill, per F8.0's week 2 Tuesday cell: **§7 Q1–10, timed at fifteen minutes.** Write the formula sheet from memory first, then attempt them — that order is the point of the exercise.